interplay is the idea of guilds, first developed by American biologist and ecologist Richard B. Root in 1967.

Root had researched the way the Blue-gray Gnatcatcher exploits its ecological niche for his doctoral thesis. The concept of ecological niches dates back to 1917, when American biologist Joseph Grinnell used the term to describe how the California Thrasher fitted into its dry, scrubby chaparral environment. The thrasher's "niche" describes the aspects of its habitat for which it is suitably adapted.

Root observed that the Blue-gray Gnatcatcher feeds on insects that live on oak leaves. By analyzing stomach contents, he showed that several other birds also consume oak-leaf insects and proposed that these oak-leaf feeding birds could be grouped into a "guild"—the "oak-foliage gleaners guild"—because they exploited the same resource.



The Blue-gray Gnatcatcher is a member of a guild of small birds that eat insects living on oak trees. Other members of the guild include Hutton's Vireo and Oak Titmouse.